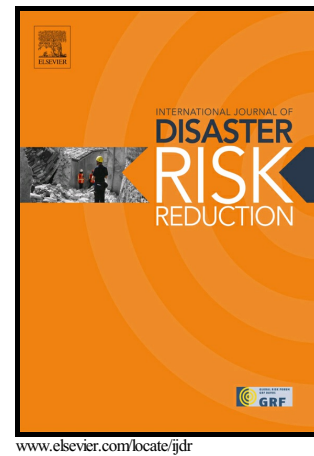


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Emergency Transportation Network Design Problem: Identification and Evaluation of Disaster Response Routes

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Abstract

Since disasters have considerable effects on transportation networks, the functionality of an emergency transportation network can play an important role in mitigation phase, especially in developing countries that sometimes suffer the sad experience of almost complete destruction of several cities. Transportation related disaster response activities typically include search and rescue, emergency medical care and fire-fighting trips. In this paper, the emergency transportation network design problem is proposed to determine the optimal network to perform emergency response trips with high priority in the aftermath of disasters. The problem has three objective functions designated to identify the optimal routes for emergency vehicles considering the length, the travel time and the number of paths, as a performance metrics of network vulnerability. A combined approach for considering the three objectives including weighted sum and lexicographic methods is used. The proposed model is solved using a branch-and-cut solution method. The suggested method is tested on the well-known Sioux-Falls network as well as on the real-world network of Tehran metropolis, Iran. Computational experiments are conducted to examine the effects of varying the maximum network length, and the relative weights of other objectives.

Keywords: Disaster mitigation, Emergency transportation network, Vulnerability analysis, Emergency response trips, Critical links.

1. Introduction

Natural disasters such as earthquake, flood, and hurricane inflict significant human and financial losses every year. The number of natural disasters and people affected by these natural disasters has increased in recent years [1]. Comparing yearly earthquake death rates among Iran, Japan and the USA during three different periods revealed that while Japan and the USA have reduced their yearly rates, Iran's status has been worsening [2]. Unlike developed countries where the main concern is to return the city to the pre-disaster conditions quickly, developing countries attempt to rescue more people in the response phase [3]. Many Iranian cities have suffered from earthquakes. Statistical analyses show that Iran has

experienced almost 130 earthquakes with magnitudes of 7.5 Richter or greater during the last century [4]. Tabas city, destroyed by an earthquake in 1978, Manjil and Rudbar cities, destroyed in 1990, and Bam, which was mostly reduced to rubble in 2003 [5]. Iran, like other developing countries, suffers earthquake casualties and economic damages. In the devastating Bam earthquake in 2003, conditions were not returned to normal and desired status in a reasonable period, e.g. within 72 hours. Following the earthquake, the airport was closed because of severe nonstructural damage suffered by the terminal and control tower [6]. In the Bam disaster, almost all of the major hospital facilities were destroyed, and a number of their doctors and nurses were seriously injured or killed. This resulted in the need to transport a great number of people to hospitals in unaffected backup areas [7]. The 26 December 2003 earthquake in Bam did not damage the roadway or the bridges in any significant manner. However, in reality, the capacity was severely lowered within the 25 kilometer vicinity of Bam. Due to the heavy influx of rescue cars and private vehicles belonging to those searching for relatives, a lengthy traffic jam occurred along the Bam-Kerman road immediately after the earthquake. After about 8 hours, the police managed to control the traffic. In Bam, civil and military aviation organizations played an important role in the rescue effort [6]. In addition, in Manjil (1990) and Bam (2003) earthquakes, nearly all narrow roads in the cities were partially or completely blocked, causing considerable delay in rescue and relief operations. Basically, most of the old urban areas in Iran suffer from narrow roads and streets [8]. Generally, natural disasters result in many devastating damages in developing countries compared to developed ones, and, therefore, make the rescue and relief operations more significant. The increase in natural disasters in the last few decades and the strong dependency of developing countries on their transportation systems has increased the need for transportation reliability studies. This necessity is mostly attributed to the dependency of an efficient transportation network to provide accessibility and support the safe and efficient movement of victims and goods [9]. In addition, if the proper paths for use of the disaster relief forces are not planned, the activity of these groups may be stopped. Without functional routes, it is impossible to respond to the affected city. Road transportation could not operate well because of the temporary traffic jams on the main roads. Prior to and immediately following the event, extricating people and providing access to health care facilities are the top priorities; therefore, the major arterial routes give priority to emergency services staff like police, fire-fighting, and ambulance services. Emergency operations infrastructure, such as the emergency operations center and supply distribution centers, are normally the next priority [10]. In addition, in practical condition, control of emergency routes is done by police forces, especially in the 72 hours after the crisis when traffic condition is not normal. Therefore, separation of routes for emergency trips with the top priorities can be effective. To manage the transportation network in the response phase, the Japan International Cooperation Agency (JICA)'s study divides the transportation network into two sub-networks, primary and secondary [11]. The emergency transportation network design problem (ETNDP) proposed in this paper seeks to identify the (primary) emergency transportation network (ETN) which is not open to public traffic. This network is controlled by police for emergency trips to preserve its role and can be used to perform disaster relief operations. It is mentioned that the allocation of some parts of network as primary network does not mean that the other parts should not be used in emergency relief operations, but simply it is called primary due to its controlling priority. Also, this does not mean that other general (non-rescue related) trips cannot use the primary network, but they only should not use those lanes of the primary network that are fully controlled and allocated to emergency response trips.

In this paper, a three-objective integer programming model for the ETNDP is presented. The ETNDP seeks minimum length of a network that minimizes the total travel time over the

selected routes and the total number of paths passing through each link as a performance metric of network vulnerability for emergency response trips. For solving the ETNDP, a combined multi-objective solution methodology using a branch-and-cut algorithm is proposed and implemented. The proposed method is tested on the well-known Sioux-Falls network and on the Tehran's network with data provided by Tehran Disaster Mitigation and Management Organization (TDMMO). The main contributions of this paper are (1) presenting a new three-objective model to identify disaster response routes, (2) developing a combined solution methodology to solve the proposed model, and (3) applying the proposed method to a real-world case study. The rest of the paper is organized as follows. A review of the related literature is presented in Section 2. The proposed model and solution method is explained in Section 3. In Section 4, results of applying the proposed method on two case studies are provided. Sections 5 and 6 provide the discussion and, the conclusion and future work.

2. Literature review

Many researchers in the late 1980s focused on operations research in disaster management. General and relatively comprehensive literature reviews in this field can be found in Altay and Green [12], Caunhye et al. [13], Galindo and Batta [14], Manopiniwes and Irohara [15], Faturechi and Miller-Hooks [16], Hoyos et al. [17], Özdamar and Ertem [18] and Zheng et al. [19]. Considering classifications of Cavdur et al. [20], the ETNDP is mainly related to the mitigation phase (pre-disaster). The pre-disaster planning stage involves infrastructure improvement to reduce vulnerability to disasters, enhancement of system resilience, and developing emergency response plans [21]. A variety of performance metrics has been proposed in the disaster literature for evaluating and analyzing disaster effects on transportation networks. These measures can be categorized generally as risk, vulnerability, reliability, robustness, flexibility, survivability, and resiliency. Other performance metrics, such as total travel time, throughput, economic loss, and connectivity provide input in quantification of some of these measures [16]. Based on the literature, connectivity, travel time of paths, and travel demand between origin-destination pairs are distinguished as highly significant factors, and the connectivity is clearly the first priority in the context of disaster mitigation [22]. Also, Sakuraba et al. [23] stated that road network accessibility becomes an important issue during the mitigation phase since a large amount of the city structures might be strongly affected, with rubble blocking several roads and access to the major distribution centers. In a similar way, in the proposed ETNDP, the connectivity is selected as a key performance measure.

Conceptually, there is a difference between the measures used for evaluating the performance of the networks subjected to disasters. de Oliveira et al. [24] has investigated reliability and vulnerability works of road networks, analyzing their similarities as well as their differences. Khademi et al. [3] has presented a list of studies for vulnerability analyses of transportation networks. Reliability is associated with the average performance in everyday situations while vulnerability (or robustness) is related to temporally random incidents that can occur. The results show that the vulnerability metrics are more strongly affected by the characteristics of alternative routes while the reliability metrics are more sensitive to the congestion level [24]. To some extent, the ETNDP can be accounted as a transportation protection problem (TPP). This problem identifies a subset of links and critical routes of the network by considering some resource limitations along with the network performance after the disaster. In fact, the TPP is a variant of the network design problem (NDP) which seeks to increase the capacity of some existing links or to add new links to network in order to improve the network reliability or resilience (see e.g. Chootinan et al. [25]; Sanchez-Silva et al. [26]; Fan and Liu, 2010; Chen et al. [27]; Shariat Mohaymany and Babaei [28], Shariat Mohaymany and Babaei [29],

and Soltani-Sobh et al. [30] which consider other aspects of the TPP in emergency situation). Most of such studies have focused on link capacity degradation in a (demand and/or supply) variability context or multiple uncertainties such as Soltani-Sobh et al. [31]. While disasters can lead to link failures and investment can enhance their survivability, the key goal of the pre-disaster investment planning problem is to identify the subset of network links to invest in under resource limitation. So, the TPP can also be defined as an important category of network design problems related to prioritizing facilities to reduce the damage caused by disaster [21, 22, 32-44]. Regardless of the disaster type, severity, losses and demand quantity, identification of critical routes can be considered as a separate research. Thus, the separation between critical routes determination and the distribution problem (i.e. mainly include victim evacuation and relief distribution problems) can be effective practically. Some distribution problems can be found in Haghani and Oh [45], Viswanath and Peeta [46], Özdamar et al. [47], [Sheu [48], Sheu [49]], Afshar and Haghani [50], Bozorgi-Amiri et al. [51], Jeong et al. [52], and Mohamadi and Yaghoubi [53]. From this aspect, the output of the ETNDP can be the input of the distribution problems.

Few works are focused on the design of emergency transportation network such as Viswanath and Peeta [46] and Shariat Mohaymany and Pirnazar [35]. Viswanath and Peeta [46] focus on identifying critical routes under disaster conditions. Viswanath and Peeta [46] developed a multi-commodity maximal covering network design problem for planning critical routes for earthquake response with two objectives: minimizing the total travel time and maximizing the total population that can be covered by these routes. The solution set that illustrates the trade-offs between the conflicting objectives for different budgets is constructed to provide practical insights to decision-makers (DMs). Shariat Mohaymany and Pirnazar [35] proposed a goal programming (GP) approach to the maximal covering network design problem (MCNDP) to identify the emergency routes.

In the TPP, even though the trip purposes and the locations of the origins and destinations of the trips are known, the number of these trips, their modes, times, and route choices that are considered as inputs are highly uncertain and, in particular, any modeling may be reasonably inaccurate. Mostly, the availability of accurate information about network disruptions when natural disasters strike is very limited. Therefore, making use of approximation or simplified procedures that are suitable for analyzing practical transportation network design problems under uncertain conditions is needed, though the simplified procedures are also useful for expediting the scheduling of response activities and controlling the network. Although the demand for all types of trips after a strong earthquake may not be derived correctly, emergency trip types and the number of paths can be predicted. The number of paths traversed considering emergency trip types through each link in related works, such as Shariat Mohaymany and Pirnazar [35], are ignored. The ETN preserve major emergency response trips through connecting major relief centers such as shelters and hospitals that are pre-located in a city and can be used to carry out logistics operations in the aftermath of disasters. The focus of the problem is ensuring appropriate connectivity for the first responders in the immediate aftermath of an earthquake. Due to a highly uncertain condition in a catastrophic disaster, as all links of the surviving ETN under disaster may fail, there would be great uncertainty in the possible closure of the transportation network links. Therefore, it may be deduced that the existence of more paths would provide a greater likelihood that the OD pair is connected and operable. On the other hand, the existence of more independent paths (with shorter travel times) will decrease the vulnerability by considering the random nature of earthquakes. Thus, in this paper, by using the vulnerability metric, the uncertainty in the connectivity of OD pairs can be considered implicitly.

3. The emergency transportation network design problem (ETNDP)

In the implementation and confirmation of the ETN, many groups such as the municipality, police, military, traffic, medical care managers are involved. Hence, there is a need to develop a decision support system (DSS) that is simple, practical and has efficient solving time. In addition, the proposed method should suggest the ETNs with different conditions to provide practical visions to decision-makers. The ETN is the most important output of the proposed model. The ETN in disaster relief operations is a higher-level network mainly composed of urban freeways and highways that are controlled by police for emergency trip sets only. Moreover, routing of emergency vehicles for each response trip set is another important output of the proposed method. In addition, a path-link performance measure of network vulnerability is introduced for the ETNDP. The proposed approach can be used for real applications in large cities and can help in prioritization of transportation network links, satisfying a certain level of redundancy or connectivity with a limited resource. In this section, first, the concepts and the methodology of the model are described, and then the model and the solution method are explained.

3.1. Concepts

When a disaster occurs, police, fire-fighting, emergency medical services, as well as emergency managers and numerous others, are involved in the response processes within the affected city. Fig.1 shows emergency trip sets connecting different points of supply and demand including the city entrance gates, railway stations, airports, *humanitarian aid distribution centers* (HADCs) supply help to the *points of demands* (PODs) directly, supported HADCs (or SHADCs), which provide educational or other facilities in non-disaster conditions, *distribution centers* (DCs), *disaster management center* (DMC) as disaster management headquarter, fire-fighting stations, Red Crescent stations, hospitals, emergency stations (ES) also known as emergency medical services (EMS), and *refugee camps* that are a temporary settlement built to receive refugees and people in disaster situations. Food, rescue materials, and other materials in major DCs are ready to be dispatched to the affected city. In an emergency logistic network, there are two major emergency trip sets: trip set of disaster relief and trip set of wounded people. It could mean wounded people who need an emergency trip to a hospital or EMS, but also transporting survivors towards *refugee camps*. Emergency trip sets can be created based on the opinion of experts. As shown in Fig.1, entrance gates to the related HADC, railway stations and airports to DCs, HADCs to the nearest HADC, HADC to the related SHADC, DCs to related HADC, DCs to the related entrance gate, DCs to DCs, DMC to DCs, DMC to HADCs, SHADC to the nearest fire-fighting station, Red Crescent stations to the nearest SHADC, hospitals to the nearest SHADC, ESs to the nearest SHADC, fire-fighting stations to the nearest SHADC, SHADCs to the nearest ES, SHADCs to the nearest Red Crescent station, SHADCs to the nearest refugee camp, refugee camps to the related entrance gate, and SHADCs to the nearest hospital are important emergency trip sets that can be considered in the ETNDP. Most of these emergency trips are done on transportation networks.

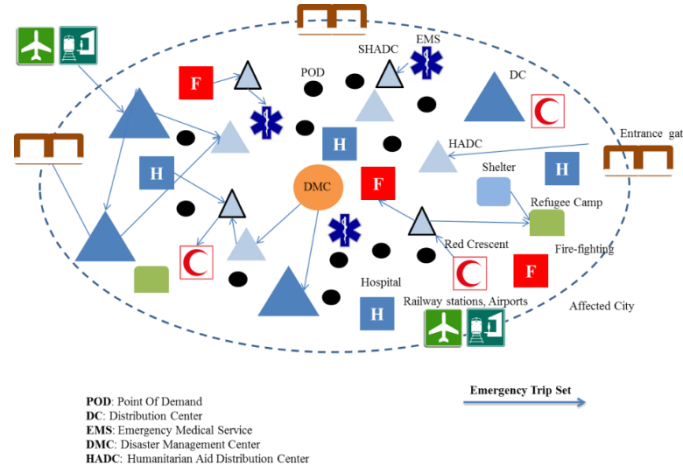


Fig.1. Emergency response trip sets network.

The ETN will support emergency operations. The 72 hours that follows the earthquake can be considered as *the emergency rescue period*, after which the chance of survival of the population may decrease dramatically [54]. The transportation network studied in this research is confined to road networks, and the effect of a disaster is investigated on the road network for the first hours immediately after the disaster strikes (e.g. the Golden 72 hours after the earthquake).

If an emergency trip has to be made only via a particular part of network (or only via special types of links such as freeways and expressways) the origin and destination of the trip should necessarily be located on that particular part of the network, which is practically impossible. Thus, all parts of the network should operate jointly for all allowed trips. On the other hand, if we want to make a network controllable under the chaotic situations happening after the earthquake (during the first three or four days), there would be no way unless using police force. Due to the limited number of relief or police forces and the limited budget for maintenance, renovation and retrofitting, the total length of the ETN should be limited as well. Therefore, to manage the transportation network in the response phase, it is reasonable to divide the transportation network into the sub-networks. In the suggested emergency road network in Istanbul, the primary, secondary and tertiary roads are determined. This order is made based on considering 1) disaster damage information collection, 2) proper emergency response operations, and 3) emergency goods circulation after the earthquake [55]. Japan International Cooperation Agency [11] has introduced two different emergency road networks in Tehran, primary and secondary, that should be controlled for the earthquake response phase (see Fig.2). This plan determines which links must be dedicated to primary network and which to the secondary one, and also specifies which trip types are allowed to use each network after the earthquake [56].

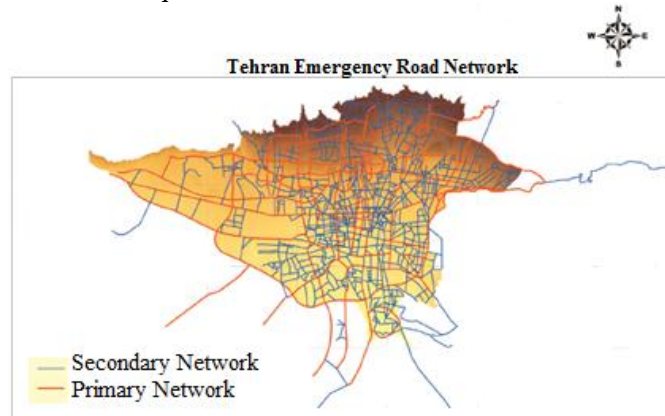


Fig.2. Tehran Emergency Road Network [11].

The *primary network* is a higher level network mainly composed of urban freeways and highways which are not open to public and is controlled by police to allow for emergency trips only. Trips made by *emergency responders* to accomplish this task are a significant part of emergency response trips loaded on transportation networks immediately after an earthquake. As stated in Rahimian [10], these trips are important because of their quantity and emergency nature. The *secondary network* is a network that is open to public traffic as well as all types of emergency trips. The primary network includes a set of *disaster response routes* (DRRs). The users are not divided into primary and secondary. Those users (vehicles) carrying *responder identification* will be allowed to access and travel on DRRs. Nonetheless, it would be necessary that the ordinary citizens who want to use DRRs must have a proper vehicle or personal identification. It should also be noted that the disaster response personnel have more responsibilities than just transporting injured people (victims), such as maintaining law and order, putting out fires, rescuing trapped people, restoring water supply, restoring electricity supply, maintaining traffic control, and treating injured people, which may be done by ordinary citizens too. During and shortly after a disaster, the ETN will support emergency operations and helps emergency responders get through to people who need help quickly. In fact, the DRRs is different from the evacuation routes [11, 56]. The DRRs have priority in clearing, repairing and restoration of all other roads. All DRRs should be clearly marked with road signs. The key to their success is public awareness of their location and knowledge of how they will be used during an emergency [57]. The proposed ETNDP is suitable to determine which links must be dedicated to the primary network. It should be noted that in the case of multi-lane DRRs, all of the lanes may not be required for the DRR, and traffic may be allowed to the extent that the responder traffic is not impeded. Also, the OD set of primary networks that is different to secondary one is considered as input. The intent of the primary network is not to prevent access by residents to their homes. By any activation of the primary network, the short-term traffic control will normally be implemented by the local police force as is supposed in the paper. Once the DRRs have been activated, the general public will be asked to keep off those lanes of routes completely or at specified times. Other lanes of the links constituting the DRRs can be assigned to the needs of the evacuees [57]. As the ETNDP aims to produce a set of disaster response routes in short-term applications (namely the emergency rescue period), we have assumed that traffic facilities should be controlled only by police force. Such situations can be imagined as normal situations where the traffic control devices within an area are broken down simultaneously and the local police force are responsible for conducting all vehicular movements at arterials intersections and/or maybe redirecting the traffic via detours. Though as stated in [57], it is also necessary to plan for longer term traffic control with non-police personnel and/or equipment. There are three types of “longer term” intersection treatments required as part of the disaster response route activation: 1. Non-movable Barricades 2. Movable Barricades 3. Traffic Control.

3.2. Methodology

Fig. 3 shows the main steps of the proposed methodology. The main characteristics of a transportation network that should be included in the data preparation phase are: road attributes, set of location of major disaster centers (MDCs) including disaster relief, shelters, and distribution centers, coverage points and other important points that are the governmental centers and agencies such as the presidential administration, the city municipalities, and the police departments. The network preparation part includes three main stages: determination of node and links (step 1), creation of connector (supplementary) links (step 2), and

determination of emergency trip sets (step 3). The optimization part includes: initialization (step 4) and development of the (three-objective) ETNDP model (step 5). Solution methodology includes: creation of emergency response paths (step 6) and solving the reduced version (single-objective) of the ETNDP (step 7). The ETN, emergency routes, and critical links are main outputs of the proposed model. The proposed model suggests the primary emergency road network which should be controlled by police to provide secure routes for the earthquake response activities.

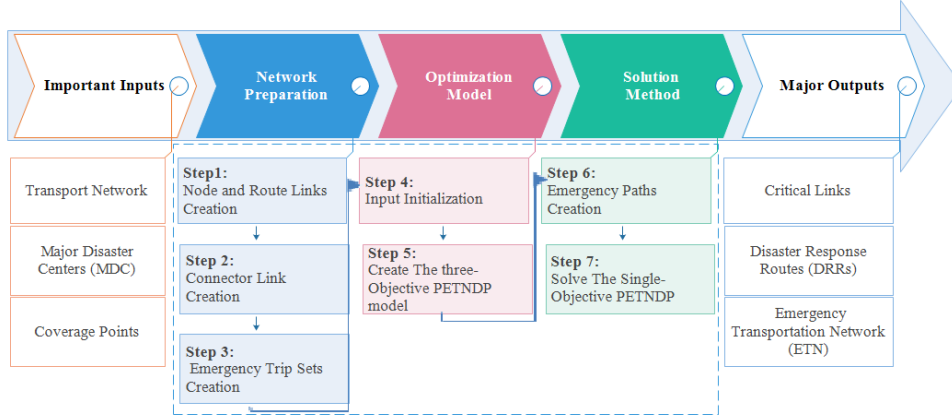


Fig.3. Main steps of the proposed methodology.

The ETNDP is the critical routes selection problem and can be viewed as a covering network design problem that attempts to find routes with the lowest travel time and vulnerability while covering the major disaster centers. In this paper, a practical optimization-based model is proposed for the determination of emergency vehicle routes for disaster response activities. In this section, first, network preparation including the inputs and nomenclature of the model parameters and variables are described, and then the proposed optimization model and its solution method are explained.

3.3. Network preparation

Let $G = (N, A)$ be a (freeway and expressway) *network* with a set of nodes N and a set of links A (Nodes and links are created with linkage of data in ArcGIS software). Each *link* $a \in A$ is characterized with a time t_a and a length l_a . It is possible that only one route or part of a route will be needed and the rest will remain available (unused) for the use of the general public. Set of nodes includes *intersection*, *mandatory access*, and *coverage* points. Set of links includes *road* and *connector* links. The *intersection* nodes are mostly intersections and roundabouts. The *mandatory access* points represent the origins or destinations of specific disaster facilities. The *mandatory access* points (such as support emergency locations and hospitals) and the *coverage* points are not directly connected to the network, but instead by means of a set of *connector* links. The *connector* links have zero travel time. Coverage of a population center is associated with the ability of a link to provide access to it. Schools and mosques are examples of the *coverage* points, where after an earthquake people may temporarily be settled in. A_k denotes set of links providing coverage to node k . An emergency *trip* corresponds to a flow that connects a pair of *mandatory access* points (From-To pairs) (*FT* set). The *OD* pairs connect *mandatory access* centers such as hospitals, fire-fighting stations, emergency centers and railway stations. For each trip set, a possible route is considered as the shortest path connecting the *OD* pair. Set of routes is shown by R_{OD} . Set of paths that include link a is denoted by R_a . The cost of each link is associated with its travel time. The travel time is calculated with an average velocity of relief vehicles, which varies for different types of road. In Table 1, the notations and required parameters for the formulation are presented.

Table 1. Notations.

Sets	
A	set of all links of transportation network, indexed by a and m
A_k	set of links providing access to node k ; $A_k \subseteq A$
N	set of all nodes, indexed by i and j
C	set of type of coverage points, indexed by c
K	set of coverage points, $k_c \in K$ (c denotes the type of k)
FT	set of From-To pairs consisting of origin and destination nodes of emergency trips, indexed by OD
R_{OD}	set of all paths connecting pair OD , indexed by r
R_a	set of paths that include link a .
Parameters	
T_r	travel time of path r
l_a	length of link a
t_a	travel time of link a
L	maximum allowable length for the ETN
R	maximum permitted number of paths traversing each link
M	a large enough positive number
N	number of alternative paths that should provide multiple connections between each OD pair
N_c	number of coverage points of type c
α_c	required coverage of points with type c
I_r	weight (or importance) of path r
w_2, w_3	relative weights of objective functions components
Variables	
y_a	1 if link a is used, and 0 otherwise
x_r	1 if path r is used, and 0 otherwise
n_a	the number of paths traversing link a
z_k	1 if point k is covered, and 0 otherwise

3.4. Optimization model

3.4.1. ETN performance metrics

Mitigating the potential impact of disaster risks requires careful identification of emergency routes. It may be important to ensure that critical links are protected so that the risk of losing an important route is substantially reduced [58]. Khademi et al. [3] suggested that under such an extreme chaotic environment after the catastrophic earthquake, where there is great uncertainty in the possible closure of the transportation network links, it may be deduced that the existence of more paths would provide a greater likelihood that the OD pair is connected and operable. On the other hand, the existence of more independent paths and paths with short travel time in the ETN decreases the vulnerability with considering random nature of earthquakes. Zhang et al. [59] stated the multiple paths will share the risk caused by the uncertain road obstruction. We chose the modified network robustness index (NRI) of Sullivan et al. [60] for the practical application, because of its ease of determination. The NRI measures how critical a link is according to its importance to the transportation network's robustness. The process to determine the NRI is based on a comparison of the total travel time of the network in situations with link m (i.e. $\sum_{a \in A} t_a n_a$) and without the link m (i.e.

$\sum_{a \in A/m} t_a^m n_a^m$). t_a^m and n_a^m denote respectively the travel time of link a without link m and the

number of paths traversing link a without link m . The NRI represents the difference in time, according to the following equation:

$$NRI_m = \sum_{a \in A/m} t_a^m n_a^m - \sum_{a \in A} t_a n_a \quad (1)$$

The network trip robustness (NTR) is calculated by summing the NRI values across all single links and dividing that sum by the total number of emergency trips in the network. The total number of OD pairs on the network for our problem is fixed. Regarding the equilibrium issue, we believe that during the emergency rescue period the equilibrium is far to be expected due to the fact that a vast number of new trip patterns will form and chaotic conditions will happen as well as that, in such cases, the travelers do not have enough time to get familiar with the degraded manner of the network as expected in long-term applications. On the other hand, as we have assumed that the ETN network is one that is under the control of police forces allowing only emergency trips, the travel times on the network's links are calculated irrespective of traffic congestion related to equilibrium issue. Since the focus of the problem is ensuring connectivity for the first responders in the immediate aftermath of an earthquake, daily traffic patterns are no longer relevant [38]. Hence, in the ETNDP, we assume that the links are not congested, and flows are completely controllable in the network to achieve a system optimum plan. Also under such non-congested condition, the response operations ensure that emergency trips can be made at free-flow travel times on the corresponding paths. By considering $t_a^m = t_a$, the NTR for our problem can be rewritten mathematically as Equation (2).

$$NTR = \frac{\sum_{m \in A} NRI_m}{\text{Total number of emergency trips}} = \frac{\sum_{a \in A/m} t_a^m n_a^m - \sum_{a \in A} t_a n_a}{\text{Total number of emergency trips}} = \frac{\sum_{m \in A} \sum_{a \in A} t_a (n_a^m - n_a)}{\text{Total number of emergency trips}} \quad (2)$$

The NTR is measured as a system-wide robustness in transportation networks with isolating links. By removing link m , the paths traversing link m will shift to other paths consisting of some other links like link a . Without link m , all paths on the network are re-routed, in a way that the number of paths traversing link a can either increase or decrease. Based on Equation (2), the lower the quantity of $(n_a^m - n_a)$ and (t_a) , the lower the link-based vulnerability imposed by link a , if there is at least a path between OD pairs. If n_m takes a small value (say μ), then the maximum possible value of other path-link variables (n_a^m) becomes $n_a + \mu$. Therefore, the difference between the traversing paths with and without link m is less than or equal to μ (i.e. $n_a^m - n_a \leq \mu$). Therefore, the ETN with minimum values of n_a and t_a is the most robust network.

The path-link value calculations in the model procedure are dependent on the number of links. Therefore, including the number of links in the denominator will better show the impact of the number of links to network robustness. For considering this issue in the ETNDP, parameter R has been introduced. R shows the maximum permitted number of paths passing each link as follows:

$$n_a \leq R, \quad \forall a \in A \quad (3)$$

By extending and summation of Equation (3) over all links, a Network-based Path-Link performance metric (NPL) can be defined which better reflects the impact of path-link variable and can be used for comparison of different ETNs. The *NPL*, as a performance metric of network vulnerability, can be written mathematically as follows:

$$\left(\frac{\sum_a n_a}{\text{Number of links}} \leq R\right) \Rightarrow NPL = \frac{\sum_a n_a}{\text{Number of links}} \quad (4)$$

The surviving network under disaster in the ETNDP is considered using NPL index via the number of paths and total time objective functions, and *R* parameter.

3.4.2. Constraints

$$\sum_r x_r = N \quad , \quad \forall OD \in FT, r \in R_{OD} \quad (5)$$

$$M \cdot y_a \geq \sum_r x_r \quad , \quad \forall a \in A, r \in R_a \quad (6)$$

$$y_a \leq \sum_r x_r \quad , \quad \forall a \in A, r \in R_a \quad (7)$$

$$M \cdot z_k \geq \sum_a y_a \quad , \quad \forall k \in K, a \in A_k \quad (8)$$

$$z_k \leq \sum_a y_a \quad , \quad \forall k \in K, a \in A_k \quad (9)$$

$$\frac{\sum_{k_c} z_k}{N_c} \geq \alpha_c \quad , \quad \forall c \in C, k_c \in K \quad (10)$$

$$n_a = \sum_r I_r x_r \quad , \quad \forall a \in A, r \in R \quad (11)$$

$$n_a \leq R \quad , \quad \forall a \in A \quad (12)$$

$$x_r, y_a, z_k \in \{0,1\} \quad , \quad \forall r \in R_{OD}, \forall a \in A, \forall k \in K \quad (13)$$

There are several paths between each *OD* pair. The idea behind the model is that there should be reliable connections between all major points. Constraint (5) is designated to allow choosing *N* of the possible connections. Constraint (6) and constraint (7) relate link and path selection variables. The path variable determines the route of the emergency vehicles and the link variable determine the selected link of the ETN. In addition, coverage constraint can be contemplated in the model for considering coverage nodes (as less important points). Coverage of a population or important center is associated with its neighbor links and can be defined as the ability of links to provide access to it. Hence, a route can cover the other important node with minimum requested coverage (α_c). Here, the types of coverage points are schools ($c=1$) and mosques ($c=2$). Coverage of ‘high importance nodes’ can be considered to be $\alpha_c=1$. Constraints (8) through (10) force the model to provide the desirable proportion of coverage for different types of coverage points. z_k equals 1 if node *k* is covered in selected link, and becomes 0 otherwise. Constraint (11) calculates n_a , which is equal to the summation over all (weighted) paths traversing link *a*, by definition. Constraint (12) controls the

maximum allowable number of paths traversing each link and distributes paths to use more links. Constraint (13) ensures the binary state of variables.

3.4.3. Objective functions

Due to the limited number of relief or police forces and the limited budget for maintenance, renovation and retrofitting that can be used for controlling and managing the network (or, generally, limited resource), the total length of the ETN should be limited. Therefore, reducing the length of the network can also be the goal of the decision maker. Let L denote the possible length of network based on available budget and/or emergency forces. Then, budget constraint can be incorporated into the problem via the total length of the network, L . In addition, the ETN must provide access to different centers with a specified quality. The quality of the ETN can be evaluated by taking into account the length, the time and the vulnerability metrics. For seeking a robust ETN, the total travel time (Time) and the total path-link (TPL) values should be minimized simultaneously as already discussed in Section 3.4.1. In addition, in the aftermath of the earthquake, the emergency response trips are sensitive to delay, and each path should have minimum time to response emergency demands. Thus, three different objective functions, Z_1 , Z_2 and Z_3 , can be introduced as follows:

$$\text{Minimize } Z = [Z_1, Z_2, Z_3] \quad (14)$$

$$Z_1 = OBJ_{LN} = \sum_{a \in A} y_a l_a,$$

$$Z_2 = OBJ_{TIME} = \sum_{r \in R_{OD}} I_r t_r x_r, \quad (15)$$

$$Z_3 = OBJ_{TPL} = \sum_{a \in A} n_a$$

The objective functions (14-15) minimize the length of the network (OBJ_{LN}), the total amount of time experienced in emergency trips (OBJ_{TIME}) and the total number of paths passing through each link (OBJ_{TPL}). As the importance of emergency trips varies for different purposes, in the OBJ_{TIME} , weighted travel times have been considered for different ODs by using I_r s. For example, trips related to victims are more important than other emergency trips.

3.5. Solution method

Consideration of more than one objective function in an optimization problem introduces additional degrees of freedom to the problem. The sorting solutions is an little importance feature when an optimization problem is single-objective since individuals are ranked only in accordance with their scalar fitness values. In the case of multi-objective optimization problems, however, this issue is more challenging because of the fitness of each solution [61]. Unless these degrees of freedom are constrained, mathematical theory indicates a set of solution points rather than a single optimal point [62]. In multi-objective programming models, solution algorithms emphasize a range of choice associated with a decision problem [63]. Several different approaches to multi-objective programming and planning problems are presented and can be found in Cohon [63] and Marler and Arora [62]. The proposed algorithm is based on the idea of a combined approach proposed by Miettinen [64]. As stated

in [62], there are many approaches to multi-objective optimization problems among which the lexicographic and weighted sum approaches have lower programming complexity, software use complexity, and computational complexity compared with other multi-objective optimization methods. Programming complexity refers to the process of programming an algorithm, whereas software use complexity concerns the actual use of an algorithm once it has been coded. Computational complexity provides a general indication of how demanding the algorithm is computationally [62]. Also, it should be noted that the efficiency of methods depends directly on the number of objective functions and constraints that is being evaluated and on the optimization engine that is being used. Hence, the suggested combined approach is a simple and practical solving methodology.

In our paper, a suggested combined approach that is a combination of the lexicographic and the weighted approaches is used. It is proven that the solution of a weighted or a lexicographic goal programming problem is Pareto optimal [64]. We have utilized this combined approach to reduce our three-objective optimization problem (ETNDP) in a single objective one. In fact, we first have reduced our three-objective optimization problem (the ETNDP) to a two-objective optimization one. Using the lexicographic goal programming approach so that, the two-objective optimization problem can be solved using the weighted sum method. In this way, the multiple objective functions are transformed into a single objective function. Then, the single-objective form of the ETNDP has been solved using the branch-and-cut algorithm.

One of the advantages of this approach is that it does not require any articulation of preferences. In no-preference methods, where the opinions of the decision maker are not taken into consideration as a priori input, the multi-objective optimization problem is solved using standard methods and then a reasonable solution is presented to the decision maker [62, 64].

In our problem, we know that the length of network is more important than the other two objectives. On the other hand, in the ETNDP, the priority of the length of network over other objectives is the basis for the proposed ETNDP. Hence, in this problem, the three objectives belong to two classes of importance. The most preferred class (i.e. the length of network) is converted into a constraint. The other two objective functions (i.e. the total travel time and the total passing path-link denoted respectively by Z_2 and Z_3) have been supposed to be equally important and belong to the same class of importance. They form the lower preference level in the lexicographic order. In the lower priority class which includes Z_2 and Z_3 , a weighted sum has been minimized subject to the most preferred class (Z_1) as constraint. Hence, the suggested combined approach is a combination of a weighted and a lexicographic goal programming problem. The suggested solution methodology can be described briefly in five steps as follows:

Step 0: Initialize

All parameters are initialized.

Step 1: Build paths

The k -shortest paths are determined in advance for the origin-destination pairs (initialize R_a , t_r). Regarding the number of feasible paths between each origin-destination pair, for practical reasons, a subset of paths using a k -shortest path algorithm has been generated to preclude the enumeration of all paths. The k -shortest paths have been created using Yen's algorithm [65].

Step 2: Build single-objective form of the ETNDP

z_2^{Min} , z_2^{Max} , z_3^{Min} , and z_3^{Max} are determined with separate runs of the ETNDP with related objective function only (initialize L , z_2^{Min} , z_2^{Max} , z_3^{Min} , and z_3^{Max}). In the lexicographic method, the objective functions are arranged in order of importance, and the most preferred

one is converted into a constraint. As the topmost goal of the ETNDP has been determined by the decision makers to be reducing the length of the ETN, Z_1 has been rewritten as a bounded constraint presented in constraint (18), which shows that the total length of the ETN should not be more than L (i.e. $(Z_1(S_1^*) + \delta_1)$). $Z_1(S_1^*)$ represents the optimum of the 1th objective function. δ_1 are positive tolerances determined by the DMs, and as they increase, the feasible region dictated by the objective functions expands [62]. Z_2 and Z_3 belong to the same class of importance and lower preference in the lexicographic order. A common approach to deal with such case of equal importance is to develop a weighted sum method as described in Marler and Arora [62]. To follow the weighted method better, Z_2 and Z_3 are normalized between 0 and 1, respectively, into $\hat{Z}_2$ and $\hat{Z}_3$ based on the most robust approach to transforming objective functions as Equations (16) and (17). Then, the single-objective form of the ETNDP with objective value (16) that is subjected to constraints (17) to (18) and (5) to (13) are solved one at a time:

$$\text{Min } F(\hat{Z}_2, \hat{Z}_3) = \text{Objective value} = w_2 \hat{Z}_2 + w_3 \hat{Z}_3 \quad (16)$$

$$\hat{Z}_2 = \frac{Z_2 - Z_2^{\text{Min}}}{Z_2^{\text{Max}} - Z_2^{\text{Min}}}, \quad \hat{Z}_3 = \frac{Z_3 - Z_3^{\text{Min}}}{Z_3^{\text{Max}} - Z_3^{\text{Min}}}, \quad (17)$$

$$Z_1 \leq L \quad (18)$$

Z_2^{Min} and Z_3^{Min} show the possible minimum value and Z_2^{Max} and Z_3^{Max} show the possible maximum value of the objective functions. To compute Z_2^{Min} , Z_3^{Min} , Z_2^{Max} and Z_3^{Max} the ETNDP should be solved by minimizing or maximizing just one of the objective functions Z_2 or Z_3 subjected to constraints (6) through (14). The solution of the suggested combined approach is Pareto optimal. This is proven in the following Theorem.

Theorem. The solution of the suggested combined approach is Pareto optimal.

Proof. The suggested combined approach is a combination of a weighted and a lexicographic goal programming problem. Our three-objective optimization problem (the ETNDP) using lexicographic goal programming problem is reduced to the two-objective optimization problem. Then the two-objective optimization problem is solved using weighted sum method (Equation (16)). So, here, only a proof for the weighted approach is needed. In [64], It is proven that the solution of a weighted goal programming problem is Pareto optimal. So, the solution of the suggested combined approach is Pareto optimal. This completes the proof. $\square$

In the case of a problem with conflicting objectives, providing the Pareto front (trade-off) curve assists DMs to select the proper strategy. The Pareto front curve was created by generating weights for problem objectives. w_2 and w_3 are positive values to weight the objective functions in order that $w_2 + w_3 = 1$. It is not practical to generate the entire solution set. Instead, it is appropriate to focus on generating an approximation to the solution set with considering the relative weights of objective functions components [30, 61]. Regarding the weight of the Z_2 and Z_3 objectives, a preliminary analysis has been performed to set parameter w_2 for different experimental scenarios so that the relative weight of objectives is set to values with 0.1 intervals.

Step 3: Solve model

Solving the model with the relative weights (w_2) is done using the branch-and-cut method. The gap tolerance is considered as a termination criterion. In most cases, the branch-and-cut

algorithm is able to solve and prove optimality for large-scale problems [66]. The branch-and-cut approach consists of a cutting plane technique embedded into a branch-and-bound framework [67]. It is a search technique that uses the tree structure in which CPLEX dynamically adds cut constraints at the tree nodes to reduce the tree size. A Java program using CPLEX software has been coded to solve the problem. The case studies are tested on a computer with Dual Core CPU at 2.8 GHz with 4 GB RAM.

Step 4: Evaluate solution

Often the decision-maker (DM) cannot concretely define what he or she prefers; this can be a difficult task for large Pareto Frontiers. In this situation, the best solution based on the metrics can be determined as described in Miettinen [64]. This approach entails selecting some solutions from a set of solutions. The solution with minimum OBJ_{LN} is considered as S^* and other solutions are denoted as S . The *Trade-off* as the solution evaluation metric represents the normalized improvement of OBJ_{Time} or OBJ_{TPL} , according to Equations (19) and (20). A trade-off reflects the ratio of change in the values of OBJ_{Time} or OBJ_{TPL} concerning the increment of OBJ_{LN} that occurs when the value of some other objective functions decrease. Solutions with larger values of *Trade-offs* are better than other solutions.

$$Trade_off (Time)(S) = \frac{(OBJ_{Time}(S^*) - OBJ_{Time}(S)) / OBJ_{Time}(S^*)}{(OBJ_{LN}(S) - OBJ_{LN}(S^*)) / OBJ_{LN}(S^*)} \quad (19)$$

$$Trade_off (TPL)(S) = \frac{(OBJ_{TPL}(S^*) - OBJ_{TPL}(S)) / OBJ_{TPL}(S^*)}{(OBJ_{LN}(S) - OBJ_{LN}(S^*)) / OBJ_{LN}(S^*)} \quad (20)$$

4. Case studies

The model performance is analyzed using two transportation networks: the well-known Sioux-Falls network, for proving the eligibility of the optimization model, to demonstrate the concepts better, and for vulnerability analysis, and the Tehran real-world network, to show the applicability of the proposed approach in modeling large-scale networks with detailed information on solution quality and efficiency. For solving the Sioux Falls network scenarios, weighted sum method is used whereas for the Tehran scenarios, the combined approach is used.

4.1. Sioux-Falls network: vulnerability analysis

The Sioux Falls network includes 24 nodes and 76 links (Fig.4). Because it is very probable that an earthquake will harm two sides of a road at the same rate, in case studies, each pair of directed links is substituted by a two-way link [68]. Therefore, the 76-link network decreases to a 38-link one. Hence, the link travel times are assumed to be symmetric. Origins and destinations of each origin-destination pair are given in Table 2. We have assumed 43 origin-destinations with different FT_IDs and $N=1$. It should be mentioned that shortest paths are created for each origin-destination pair.

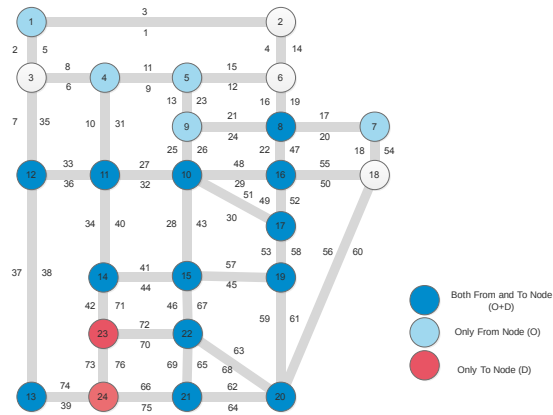


Fig.4. Sioux Falls network.

Table 2. Set of From-To pairs (FT).

FT_ID	O	D	FT_ID	O	D	FT_ID	O	D
1	1	10	16	11	13	31	15	20
2	4	10	17	11	14	32	15	23
3	4	11	18	11	15	33	16	19
4	5	10	19	11	16	34	16	20
5	7	8	20	11	17	35	16	22
6	7	16	21	11	22	36	17	19
7	7	17	22	11	23	37	17	20
8	8	10	23	12	13	38	17	22
9	8	17	24	13	22	39	19	20
10	9	11	25	14	15	40	19	22
11	9	16	26	14	22	41	20	21
12	10	19	27	14	23	42	21	22
13	10	21	28	9	15	43	22	24
14	10	23	29	15	16			
15	11	12	30	15	17			

4.1.1. Objective 1: length of network

For calculating the minimum length of the network connecting all *OD* pairs, the model was only run with respect to OBJ_{LN} resulting in 113.28 km of roads ($w_2=0$ and $w_3=0$). The length of

the network is optimized by considering OBJ_{LN} as the objective function. In this case, total travel time was 1116 seconds and 355 paths passed through all links. This network with minimum length (about 113 km) was best for the retrofit program at a first stage because it requires a minimum budget, though probably results in worst performance of the emergency response activities. Attributes of this run are as follows: number of Links = 21 (number), $OBJ_{LN} = 113.28$ (km), $OBJ_{TIME} = 1116$ (seconds), $OBJ_{TPL} = 355$ (number), and $NPL=16.90$.

4.1.2. Objective 2: total travel time

In this section, only total emergency travel time is considered as a single objective function. With $w_2=1$ and without any limit on network length L , the model has been run with R between 7 and 14 (R shows the maximum permitted number of paths passing through each link). No feasible solution for $R < 7$ is obtained. In other words, we cannot design an ETN with fewer than 6 paths passing through each link. In Fig.5, the total length of the network was between 198 and 228. The objective function, minimum total travel time, was 647 seconds. Results for three different ranges of R (i.e. $7 \leq R \leq 9$, $10 \leq R \leq 13$, and $R > 14$) can be seen in Fig.6. For R between 7 and 9, by increasing the total travel time, the total number of path-link is also increased. However, for $R \geq 9$ the trend of the objective function is different. As R exceeds 9 the total length of the network remains fixed at about 198 (but with different performance objectives). The lowest NPL is 5.34 paths per link. The overall trends of the total length of network and the total travel time are depicted respectively in Figs.5 and 6.

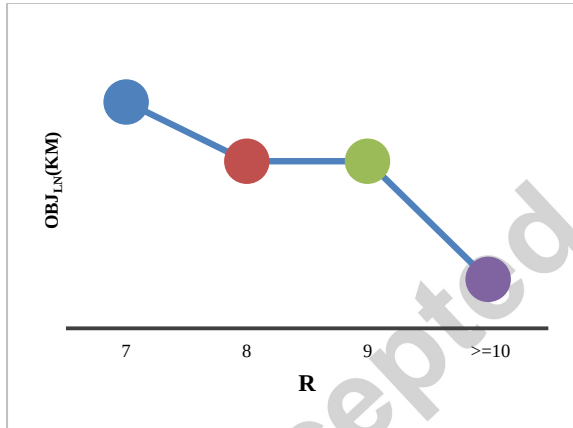


Fig.5. Evaluation of length of network (OBJ_{LN}).

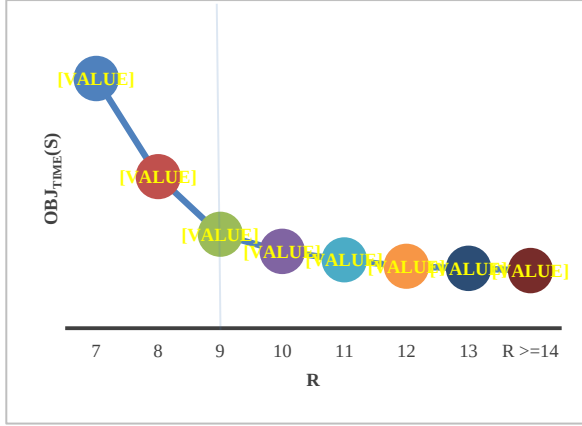


Fig.6. Evaluation of total travel time (OBJ_{TIME}).

4.1.3. Objective 3: total passing path-link

With $R=9$, the ETN has the lowest NPL with 36 links. As shown in Fig.7, for $w_2=0$ (only total path-link objective function and without any limit on network length L), by increasing R the total path-link decreases. Results show that network with a same total path-link can have a different travel time between 684 and 740 (see Fig.8).

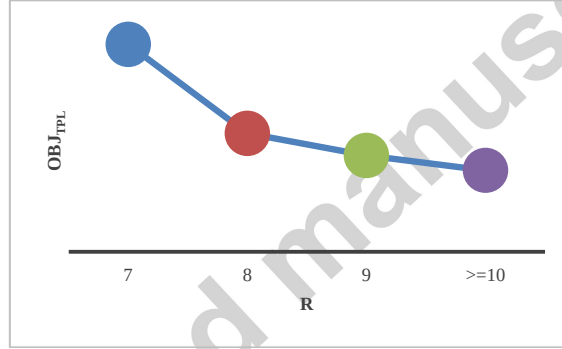


Fig.7. Evaluation of total passing path-link (OBJ_{TPL}).

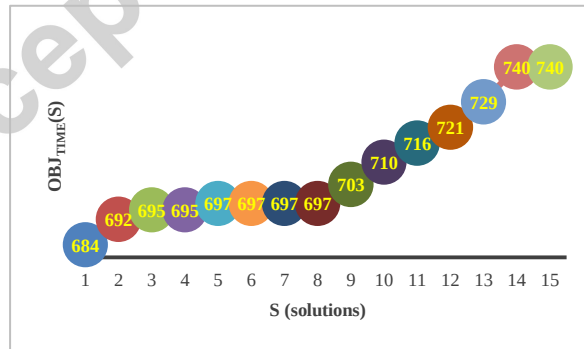


Fig.8. Comparison of networks with fixed $OBJ_{TPL} = 166$.

4.1.4. Objective 2 and 3: total travel time and passing path-link

Description for $w_2=1$ (only total travel time objective function) and for $w_2=0$ (only total path-link objective function) was presented in the previous sub-sections. For $w_2=0.5$ and without any limit on network length L and R , the initialization are done with $Z_2^{Min} = 647$, $Z_2^{Max} = 22088$,

$Z_3^{Min} = 166$ and $Z_3^{Max} = 5247$. For different values of w_2 , results are shown in Fig.9. OBJ_{Time} and OBJ_{TPL} behave reversely.

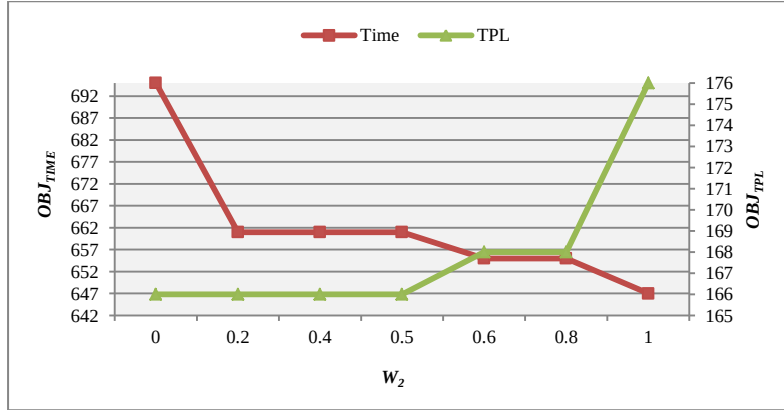


Fig.9. The performance of the ETNs with different relative weight (w_2).

In Table 3, the number of alternative paths is evaluated ($w_2=0.5$). As shown in Table.3, the total length of network increased with increasing N . If three paths ($N=3$) that should provide multiple connections between each OD pair are considered, 36 links should be used. All links of the network should be used in the case of $N \geq 4$.

Table.3. Evaluation of the number of alternative paths.

N	1	2	3	4	5
Number of links	31	34	36	38	38
$OBJ_{LN}(KM)$	214.91	238.24	248.234	266.56	266.56

For evaluating the network, first, the minimum length of the ETN to connect all OD pairs is determined. Then, the performance of the network is evaluated by considering: only the length of the network, only the total travel time, only the total passing path-link, and both total travel time and total passing path-link. Based on Table 4, the NPL value decreases from 16.9 to about 5 paths per link with considering the total path-link (TPL) only, which has the minimum total number of paths. The results of suggested model are better than what can be achieved without it.

Table 4. Evaluation of objectives.

Objective	Z_1 : length of network	Z_2 : total travel time	Z_2 : total travel time) and Z_3 : total passing path-link	Z_3 : total passing path-link
Number of links	21	30	31	32
$OBJ_{LN}(KM)$	113.28	198.254	214.91	228.24
OBJ_{TIME} (S)	1116	647	661	695
OBJ_{TPL} (Number of paths)	355	176	166	166
NPL	16.9	5.87	5.35	5.19

4.2.Case study of Tehran: identification of emergency transportation network

4.2.1. Inputs

Some Iranian cities are much more susceptible to seismic hazard due to their geological and morphological characteristics and among them, Tehran is one of the most-exposed cities. Tehran is the capital of Iran and also the largest city in western Asia and the 5th-largest city globally. This city is located in the north of the central plateau of Iran. Tehran is the most

populated city in Iran with 8,300,000 inhabitants and about 12 million in metropolitan areas in 2014, which extends about 730 km². The capital of Iran is located at the foot of the Alborz mountains range in a highly earthquake-prone area bounded by major active faults such as Mosha, Niavaran, Tarasht, and Parchin faults. Rapid urbanization and lack of strong regulations for construction can be seen in some parts of Tehran. In some parts of Tehran, especially in central and southern districts, some blocks are categorized as high priority for rehabilitation because of narrow roads. The Tehran metropolises are equipped with a huge network of highways and interchanges. The Tehran transportation network has a structure with 540 km of freeways, 776 km arterial roads, and 1468 km collectors and local streets. Local streets and some secondary arterials should be removed from the input network. The presented study considered a subset of the Tehran road network, with arterial roads and freeways with 2992 two-way links (Fig.10).

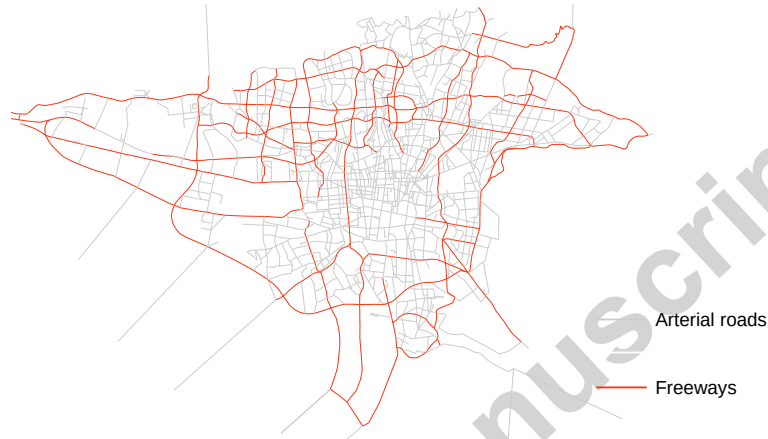


Fig.10. Tehran arterial and freeways network.

6123 coverage points including schools, mosques and other important points are considered. In this case, $\alpha_c = 0.6$ for all c , $I_r = 1$, and $N=1$. Some important points belonging to the FT set are shown in Fig.11. Based on the experts' opinion, 19 emergency trip sets are created (see descriptions presented in Section 3.1). The ETN was determined by assuming different values for L and R , or by fixing L or R to some constants. We need an upper and a lower bound on the objectives Z_2 and Z_3 by separate runs of the problem with respect to each objective function, $z_2^{Min}, z_2^{Max}, z_3^{Min}$ and z_3^{Max} values are obtained for the parameters considering separate runs. As mentioned in Section 3.5, the notion of an optimal solution is not relevant here and is replaced by the concept of a solution set from which the DMs select the most preferred solution.

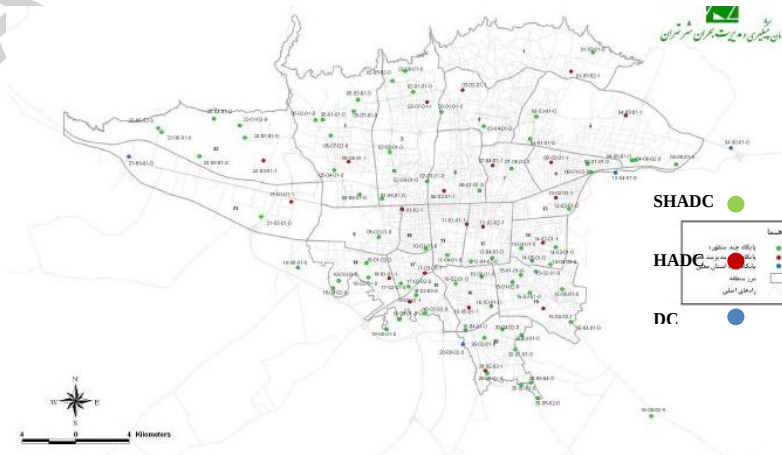


Fig.11. The location of SHADCs, HADCs, and DCs of Tehran (source: [69]).

4.2.2. The ETNs for Tehran

In order to reduce the vulnerability of network, changes in parameter R can be evaluated. For $R < 107$, there is no feasible solution. In two cases of no limitation on R and $R = 117$ solutions are the same. The shortest length for the network is approximately 823 Km.

As shown in Table 5, 11 different weights of w_2 have been considered. By considering $L = 900$, $R = 107$, the model was run for arterial and freeways network of Tehran. 823 Km is the minimum length of the network obtained for $S(w_2, w_3) = S(0.9, 0.1)$ and $S(1.0, 0.0)$ when the path-link variable is not considered. The network $S(0.0, 1.0)$ with length about 900 Km, that is the maximum possible length, has the minimum total path-link. The network has 1536 links. The NPL value of suggested network is equal to 11.72 paths per link. If the same importance is considered for OBJ_{Time} and OBJ_{TPL} , network with about 824 Km is the best network. The $S(0.5, 0.5)$ has the best *Trade_off (TPL)* considering the same weight for w_2 and w_3 . By increasing w_2 , lengths of network and total time decreased, but total path-link increased. The number of links of $S(0.0, 1.0)$ is more than that of $S(1.0, 0.0)$. The range of CPU times obtained under w_2 varied from 20.8 to 1537 seconds. By increasing importance of OBJ_{TPL} , the time increases compared with when omitting OBJ_{Time} due to ignoring the path variable (x_r). By increasing the length of network, the total emergency travel time increases but the total passing paths of total links decreases compared with when OBJ_{TPM} is omitted and the length of network is set to be equal to its maximum value (i.e. 900 KM). Emergency transportation network with length 823 has been total minimum emergency travel time (125 h) and maximum total passing paths. As shown in Table 5, $S(0.5, 0.5)$, $S(0.1, 0.9)$, $S(0.3, 0.7)$, and $S(0.6, 0.4)$ have maximum *trade-offs*. Since links with smaller n_a are better than other links in general in the vulnerability view, networks with lower R , OBJ_{Time} , and OBJ_{LN} are more preferred than others.

Table 5. Results with considering fixed R .

(w_2, w_3)	$OBJ_{LN}(KM)$	OBJ_{Time} (Hour)	OBJ_{TPL}	Normalized OBJ Value	Solving Time (Seconds)	<i>Trade_off</i> (Time)	<i>Trade_off</i> (TPL)
(0.0, 1.0)	899.777	127.61	17119	0.0000	1536.797	-0.2629	0.5467
(0.1, 0.9)	844.571	125.131	17419	0.0061	81.293	-0.1751	1.3212
(0.2, 0.8)	838.649	124.639	17827	0.0067	43.286	-0.0310	0.6181
(0.3, 0.7)	829.953	124.604	17888	0.0062	26.076	-0.0367	1.0067
(0.4, 0.6)	827.454	124.574	17981	0.0055	35.413	-0.0111	0.5916
(0.5, 0.5)	824.299	124.571	17996	0.0047	36.408	-0.0261	1.7592
(0.6, 0.4)	824.864	124.568	18012	0.0038	33.656	-0.0042	0.6658
(0.7, 0.3)	824.864	124.568	18012	0.0028	38.823	-0.0042	0.6658
(0.8, 0.2)	823.287	124.567	18033	0.0019	32.901	0.0000	0.0001
(0.9, 0.1)	823.287	124.567	18035	0.0009	30.928	0.0000	0.0000
S*(1.0, 0.0)	823.287	124.567	18035	0.0000	20.793	-	-

4.2.3. Comparison of the ETNs

For values less than 700 KM L , there are not any feasible solutions ($Z_1(S_1^*) = 700$). In two cases of no limitation on L and $L = 900$ solutions are the same. Thus, L is set for the length range 700 to 900 with 50 KM intervals ($\delta_1 = 50$) that may have significant impact on the solution set. The various instances for $L = \{900, 850, 800, 750, 700\}$ and the weight

$w_2=\{0.0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1\}$ are solved. Results are shown in Table 6 and show the trade-offs between the conflicting objectives for different maximum network lengths to provide practical visions to DMs. The maximum CPU times for the instances solved under $w_2=0.9$ and $w_2=0.8$ are approximately 1.5 and 1 minute, respectively. For $S(900,1)$ because of no control of the total number of paths (OBJ_{TPL}) and total network length (OBJ_{LN}), the network has the highest value of OBJ_{TPL} . This network has the lowest time and 823 KM length. $S(900,0)$ is the largest network and the lowest OBJ_{TPL} . The ETNs with $w_2=0.1$ have highest *trade-offs*. The suggested ETNs (solutions) based on the *trade-offs* and for $L > 700$ are shown in Fig.12. Each point on Fig.12 represents a different solution. The network for $S(L, w_2) = (700, 0.1)$, has maximum *Trade_off (TPL)* and can be considered as the candidate ETNs. The best ETNs with considering L have maximum *trade-offs* value for $w_2 < 0.3$. The results of the *trade-offs* help DMs to select the best ETNs. Fig.13 plots the L (in KM) and the objective value ($F(\hat{Z}_2, \hat{Z}_3)$). The larger networks are more desirable than smaller one from the point of vulnerability.

Table 6. Tehran emergency transportation networks: the solution set.

W ₂												
R=Unlimited	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	Average
L=900												
Solving time (s)	3.643	12.79	6.791	5.787	11.558	5.778	5.711	13.755	10.375	15.009	6.542	8.885
Objective value	0.000	0.006	0.006	0.006	0.005	0.004	0.003	0.002	0.001	0.000	0.00	0.003
OBJ _{LN} (KM)	877.275	839.377	834.838	828.433	827.453	824.298	824.863	824.230	823.286	823.286	823.286	831.875
OBJ _{TIME} (Hour)	127.390	125.120	124.634	124.583	124.552	124.548	124.546	124.546	124.545	124.544	124.544	124.868
OBJ _{TPL}	17093	17400	17806	17893	17991	18006	18022	18023	18039	18045	208896	35201.27 3
L=850												
Solving time (s)	13.739	6.693	7.014	7.92	6.87	8.866	5.675	5.705	5.742	5.849	5.72	7.254
Objective value	0.000	0.006	0.008	0.008	0.008	0.008	0.009	0.008	0.008	0.008	0.008	0.007
OBJ _{LN} (KM)	849.008	844.791	837.435	829.803	829.930	830.163	826.985	827.453	824.298	824.298	824.298	831.679
OBJ _{TIME} (Hour)	127.455	125.164	124.677	124.595	124.587	124.582	124.566	124.552	124.550	124.548	124.548	124.893
OBJ _{TPL}	17093	17374	17750	17868	17883	17896	17942	17991	17999	18006	18006	17800.72 7
L=800												
Solving time (s)	9.321	51.745	37.372	35.895	36.935	29.185	52.075	41.229	20.898	28.954	30.672	34.025
Objective value	0.000	0.006	0.008	0.009	0.009	0.009	0.009	0.009	0.009	0.009	0.009	0.008
OBJ _{LN} (KM)	799.884	799.990	799.831	799.985	799.950	799.983	799.967	799.961	799.961	799.669	799.669	799.895
OBJ _{TIME} (Hour)	127.424	125.140	124.701	124.615	124.599	124.576	124.574	124.572	124.572	124.566	124.566	124.900
OBJ _{TPL}	17132	17414	17754	17874	17905	17961	17970	17978	17978	18002	18002	17815.45 5
L=750												
Solving time (s)	45.425	46.571	22.654	11.453	8.166	14.654	23.993	20.024	10.68	8.093	13.841	20.505
Objective value	0.001	0.008	0.010	0.011	0.012	0.013	0.014	0.014	0.015	0.016	0.016	0.012
OBJ _{LN} (KM)	749.991	749.981	749.966	749.979	749.921	749.921	749.984	749.982	749.982	749.982	749.982	749.970
OBJ _{TIME} (Hour)	127.208	125.255	124.768	124.735	124.694	124.694	124.687	124.682	124.682	124.681	124.681	124.979

OBJ _{TPL}	17256	17498	17873	17924	17997	17997	18017	18037	18037	18042	18042	17883.63 6
L=700												
Solving time (s)	33.061	37.409	10.016	15.878	42.49	46.18	52.967	41.672	54.212	96.54 1	30.861	41.935
Objective value	0.004	0.013	0.017	0.020	0.023	0.029	0.032	0.032	0.035	0.038	0.041	0.026
OBJ _{LN} (KM)	699.989	699.983	699.985	699.985	699.994	699.937	699.937	699.937	699.937	699.937	699.937(S*)	699.96 0
OBJ _{TIME} (Hour)	127.410	125.556	125.130	125.130	125.084	125.077	125.075	125.075	125.075	125.075	125.075	125.34 2
OBJ _{TPL}	17538	17806	18111	18111	18194	18216	18225	18225	18225	18225	18225	18100. 091

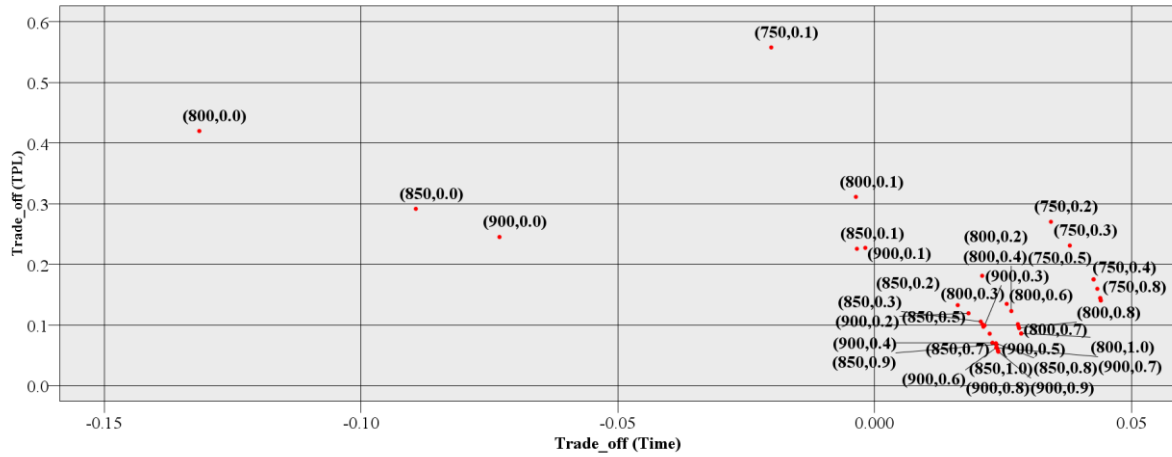


Fig.12. The solution set for different length values ($L > 700$).

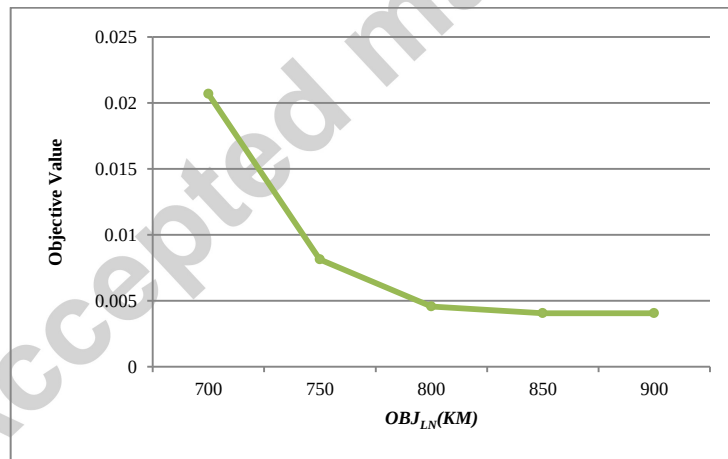


Fig.13. The objective values with different L for $w_2=0.5$.

In Table.6, with considering the total passing path-link objective, the result is better than the recent methods in the TPP that is considered the total travel time objective only. For example, for $L = 900$ and $w_2=0.3$, with decreasing 150 paths, the NPL value, about 1 percent is improved with minimum increase of total network length. The ETN is not implemented in Tehran so far. The previously suggested networks is based on expert's judgments and decisions. In our paper, the effective parameters in the determination of the ETNS are presented and are modeled as a integrated multi-objective programming.

5. Discussion

Based on Tables 4, the *NPL* value is decreased from 16.9 to about 5 paths per link with considering the total path-link and the total travel time. As shown in Table 4, with an insignificant increase of the length of the ETN, we can have a network with lower vulnerability. It is more important to consider the *NPL* in situations that the failure probabilities of links are more unpredictable.

The value of path-link variable shows the number of paths that pass through each link. The path-link values can be used for prioritizing the links. The ETNDP can be used to identify and rank the most critical links in a network using the method proposed by [60]. In addition, increasing *N* can rank critical links. This analysis of evaluation of critical links and their removal by changing parameters' values can be vital for decision makers to identify the most vulnerable links throughout the network.

As can be seen in Table 5, the *OBJ_{TPL}* values of the ETNs related to $S(0.6,0.4)$ and $S(0.7,0.3)$ with minimum increasing of the total travel time are increased. The links of these ETNs because of considering *R*, are lower than the link-based vulnerability compared with $S(L=900, w_2)$ in Table 6. The same physical network may have different objective function values.

The results show that considering the total travel time and the total path-link was very costly and led to increasing the length of network. By evaluating the average times of trip sets, relocation of disaster centers can be evaluated. In addition, the impact of the ordinary movements on the congestion of the primary network can be taken into account through the travel time of each link (i.e. t_a), which is an input parameter in the proposed model.

We have utilized a suggested combined approach that is a combination of the lexicographic and the weighted approaches to reducing our three-objective optimization problem (ETNDP) into a single objective one. The suggested solving methodology has been reduced the decision maker (DM) role in the determination of the solutions for the multi-objectives optimization problem. The programming complexity, software use complexity, and computational complexity of the combined approach are minimum compared to other approaches. Hence, the suggested combined approach is a simple and practical solving methodology. In addition, at the end of a multi-objective optimization, the DM has to select the preferred solutions from the Pareto Frontier; this can be a difficult task for large Pareto Frontiers. For this reason, the trade-offs as the solution evaluation metric have been developed to aid the DM in selecting the preferred solutions. It should be noted that the solution method described here does not require any articulation of preferences (i.e. called 'no-preference method').

There are high uncertainties in links failure probability in Tehran. When an earthquake disaster occurs, each link can be non-functional. When links of a transportation network fail because of the great earthquake, the connectivity of the network can be damaged and trip sets demand cannot be served. The proposed planning focuses on determining the links of the emergency transportation network under the uncertainty of available links after an earthquake. With using the ETNDP, the network link-based vulnerability under a disaster are evaluated. The existence of more independent paths will decrease the vulnerability by considering the random nature of earthquakes. So, If there is significant uncertainty in the probability of link failure due to the occurrence of a disaster, such as massive earthquakes, the proposed model can be useful.

6. Conclusion and future work

This paper has proposed a model for the emergency transportation network design problem that is useful in developing pre-disaster plans for controlling the network in order to carry out emergency response activities after an earthquake. The model can also be used for identifying critical routes. The problem has been formulated as a three-objective integer programming

and has been solved using a combined approach as a method with no articulation of preferences methods in multi-objective programming. In addition, the path-link performance metric for analysis of network vulnerability and solution evaluation metric for ranking solutions have been suggested. A network analysis is performed to provide the decision-maker an alternative measure for the ability of the critical routes to remain functional after an earthquake. As expected, by increasing the total travel time, the total number of the path-link is also decreased. In addition, the performance of the same physical network may be different with considering different routing of emergency trip sets. The results showed that for the first hours immediately after the disaster event, although total travel time is crucial, the number of emergency trips passing each link is also important. This is because entering extra trips on each link can increase the vulnerability of the emergency network. The results also showed the importance of considering path-link performance metric of network vulnerability in identifying primary emergency routes. The proposed model can be applicable for other types of natural disaster.

This manuscript is advanced our knowledge of modeling large-scale emergency networks. We suggest that the proposed model can be used for models related to response disaster phase such as distribution and TPP problems. The maximum permitted number of paths traversing each link can be considered for each link separately. For future study, developing the ETNDP considering retrofitting strategy and earthquake scenarios simultaneously, and considering both the emergency crew response and the individual evacuees simultaneously in the design of emergency network can be considered as a good topic for future research . In addition, considering relocation of disaster-related centers in the model can be considered. The suggested solving methodology does not generate all solution sets by considering the relative weights of objective functions components. In addition, the same weaknesses in connection with the weighted sum method and the lexicographic approach are valid for the suggested combined approach. The exact methods such as the combined approach have minimum parameters that should be tuned in comparison to the metaheuristic algorithms such as genetic algorithms (GAs). In addition, in practical problems with many constraints, the procedure of considering the constraints in metaheuristic algorithms is challenging One direction for future research in solution methodology such as our suggested method is the development of a multi-objective solution method for real case problems with the target of reducing DM role in the determination of the solutions.

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